

experiments. (iii) Training from scratch: it should offer scalable structural data to train LLMs from scratch, mitigating the data leakage and providing clean evaluations.

Additionally, our proposed framework enables the study of why and when CoT reasoning succeeds or fails.

B. Illustration of DataAlchemy Components

B.1. Elements

As defined in Section 4.1, an element e is defined as an ordered sequence of atoms. The following is an element with 3 atoms:

```
A P P
```

The length of the elements can vary and the following is an element with 4 atoms:

```
A P P L
```

B.2. Transformations

A transformation is defined as an operator acting on the elements, as detailed in Section 4.2. In this work, we employ three distinct transformations, denoted as f_1 , f_2 , and f_3 .

Specifically, f_1 represents an element-wise ROT-13 operation applied to the alphabet. The following example demonstrates the application of f_1 to the input APPL:

```
A P P L [F1] <answer> N C C Y
```

f_2 denotes a cyclic positional shift by one step. The following example illustrates the result of applying f_2 to the sequence APPL:

```
A P P L [F2] <answer> P P L A
```

f_3 denotes the sequence reversal operator. The following example displays the result of applying f_3 to APPL:

```
A P P L [F3] <answer> L P P A
```

B.3. Compositional Transformations

Compositional transformations refer to imposing multiple transformations sequentially to an element. Below are the examples of transformations f_1 and f_2 :

```
A P P L [F1] [F2] <think> N C C Y [F2] <answer> C C Y N
```

C. Illustration of Generalization Tasks

C.1. Illustration of Task Generalization

As shown in Fig. 2, task is defined through element and transformation and hence the generalization of tasks can be decomposed to generalization on element and on transformation respectfully.

C.1.1. Transformation Generalization

We consider the transformation generalization on four aspects: in distribution (ID); compositional (CMP), partially out of distribution (POOD) and out of distribution (OOD). We use the element APPL to further demonstrate the four aspects in detail:

In distribution refers to the scenario that the test transformations are identical to the training ones:

```
Training:
A P P L [F1] [F2] <think> N C C Y [F2] <answer> C C Y N
Test:
A P P L [F1] [F2] <think>
```

Compositional refers to the scenario where the test transformations are the composition of the training ones

```
Training:
A P P L [F1] [F2] <think> N C C Y [F2] <answer> C C Y N
A P P L [F2] [F1] <think> P P L A [F1] <answer> C C Y N
A P P L [F2] [F2] <think> P P L A [F2] <answer> P L A P
Test:
A P P L [F1] [F1] <think>
```

Partially out of distribution refers to the scenario where part of the compositional test transformations are seen during training while the entire compositional transformation is different from training

```
Training:
A P P L [F1] [F1] <think> N C C Y [F2] <answer> A P P L
Test:
A P P L [F1] [F2] <think>
```

Out of distribution refers to the scenario where none of the compositional test transformations are seen during training.

```
Training:
A P P L [F2] [F2] <think> P P L A [F2] <answer> P L A P
Test:
A P P L [F1] [F1] <think>
```

C.1.2. Element Generalization

We consider the transformation generalization on three aspects: in distribution (ID); compositional (CMP) and out of distribution (OOD). We use the element APPL to further demonstrate the three aspects in detail:

In distribution refers to the scenario that the test elements are identical to the training ones:

```
Training:
A P P L [F1] [F2] <think> N C C Y [F2] <answer> C C Y N
Test:
A P P L [F1] [F2] <think>
```

Compositional refers to the scenario where the test elements have the same atoms as the training ones but with a different order

```
Training:
A P P L [F1] [F2] <think> N C C Y [F2] <answer> C C Y N
Test:
P A L P [F1] [F2] <think>
```

Out of distribution refers to the scenario where the novel atoms show in the test elements

```
Training:
A P P L [F1] [F2] <think> N C C Y [F2] <answer> C C Y N
Test:
A P P Y [F1] [F2] <think>
```

C.2. Illustration of Length Generalization

Similar to above, we decompose the length generalization into text length generalization and reasoning step generalization.

C.2.1. Illustration of Text Length Generalization

Text length generalization involves different lengths of elements in the test set from those in the training set. Still, we use APPL as an example to illustrate text length generalization.

```
Training:
A P P L [F2] [F2] <think> P P L A [F2] <answer> P L A P
Test:
A P P L E [F1] [F2] <think>
Test:
A P P [F1] [F2] <think>
```